WATERWATCH · DESIGN PRECIS

WaterWatch design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.9	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25: concept A chosen (O1); photometer as the monthly reference (O2); R1 restated above 1.0 mg/L (O5); drift room 0.15 mg/L
0.8	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: describes option A of WWT-OPT-001 (proposed, awaiting Amish); battery is one 32700 cell (WWT-DEC-001 F1, after STEM check S3); requirement numbers now PR-001 to PR-016 (WWT-PRS-001)
0.7	2026-10-02	Amish Chadha	Energy, autonomy and cost from WWT-CAL-001 v0.4 (status light, WWT-DEC-001)
0.6	2026-10-02	Amish Chadha	Partner, alert and data open questions answered by Amish's decisions of 2026-10-02 (WWT-DEC-001)
0.5	2026-10-02	Amish Chadha	Design made constructable (WWT-DDR-003); components table, cost and drawings updated; budget treated as a value-engineering target
0.4	2026-09-25	Amish Chadha	

			Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	For TRL 3, decisions from WWT-DDR-001; numbers from WWT-CAL-001; smaller flow cell, pressure-compensating regulator, 30 s settling wait, air-break drain and sun shield; parametric model and drawing WWT-DWG-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, components, first-order numbers, safety, open questions, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

WaterWatch design precis

WaterWatch is a solar sentinel that stands on its own pole beside a village tapstand. Once an hour a latching valve on a tee in the tapstand riser opens for 90 s and flushes water through a small, dark flow-through cell. An infrared nephelometer measures turbidity, a membrane-free three-electrode sensor measures free chlorine and a sealed probe measures temperature. An ESP32 controller logs the readings, a cellular modem uploads them every 4 h, and an SMS goes to the caretaker and operator when chlorine drops below 0.2 mg/L or turbidity passes 5 NTU. The TRL 3 calculations (WWT-CAL-001) give 0.54 Wh per day, 28.5 days on a 3.2 V, 6 Ah LiFePO4 cell without sun, 18.0 L per day of flushed water and $325 in parts for the constructable design (WWT-DDR-003 and the status light of WWT-DEC-001), USD 25 over the USD 300 value-engineering target. How each part is made and fitted is in the prototype build plan (WWT-BLD-001). The weak point is still the chlorine reading. Under Amish's decisions of 2026-09-25 (WWT-DDR-002), R1 is relaxed to ±0.2 mg/L or ±25 %, and sites whose pH is above 7.5 or moves more than 0.2 between visits take a pH probe in a spare port of the cell. With that rule the error budget of WWT-CAL-001 v0.5 met R1 before sensor drift; the independent review of the STEM check corrected the budget, which then misses R1 at 1.5 mg/L by up to 0.017 mg/L, so R1 is not met there (a restated target above 1.0 mg/L is proposed, awaiting Amish, WWT-DEC-001 O5), while it is met with room at 0.2 to 0.5 mg/L where the alerts are set. The low-cost sensor's drift over months is unknown, so R10 remains at risk. The design choices below were decided by Amish on 2026-09-25 (WWT-DDR-001 and WWT-DDR-002).

This precis describes option A of the concept options (WWT-OPT-001): the design as it stood before kit 1.8, scored against two alternatives with different working principles, a reagent colorimetric analyser (option B) and probes in the riser (option C). Option A scores highest and is recommended; the choice is proposed, awaiting Amish (WWT-DDR-004). The STEM check (WWT-STM-001) found one detail to correct, the battery's cell count, now one 32700 cell (WWT-DEC-001 F1). The requirements R1 to R13 named below are PR-001 to PR-013 in WWT-PRS-001.

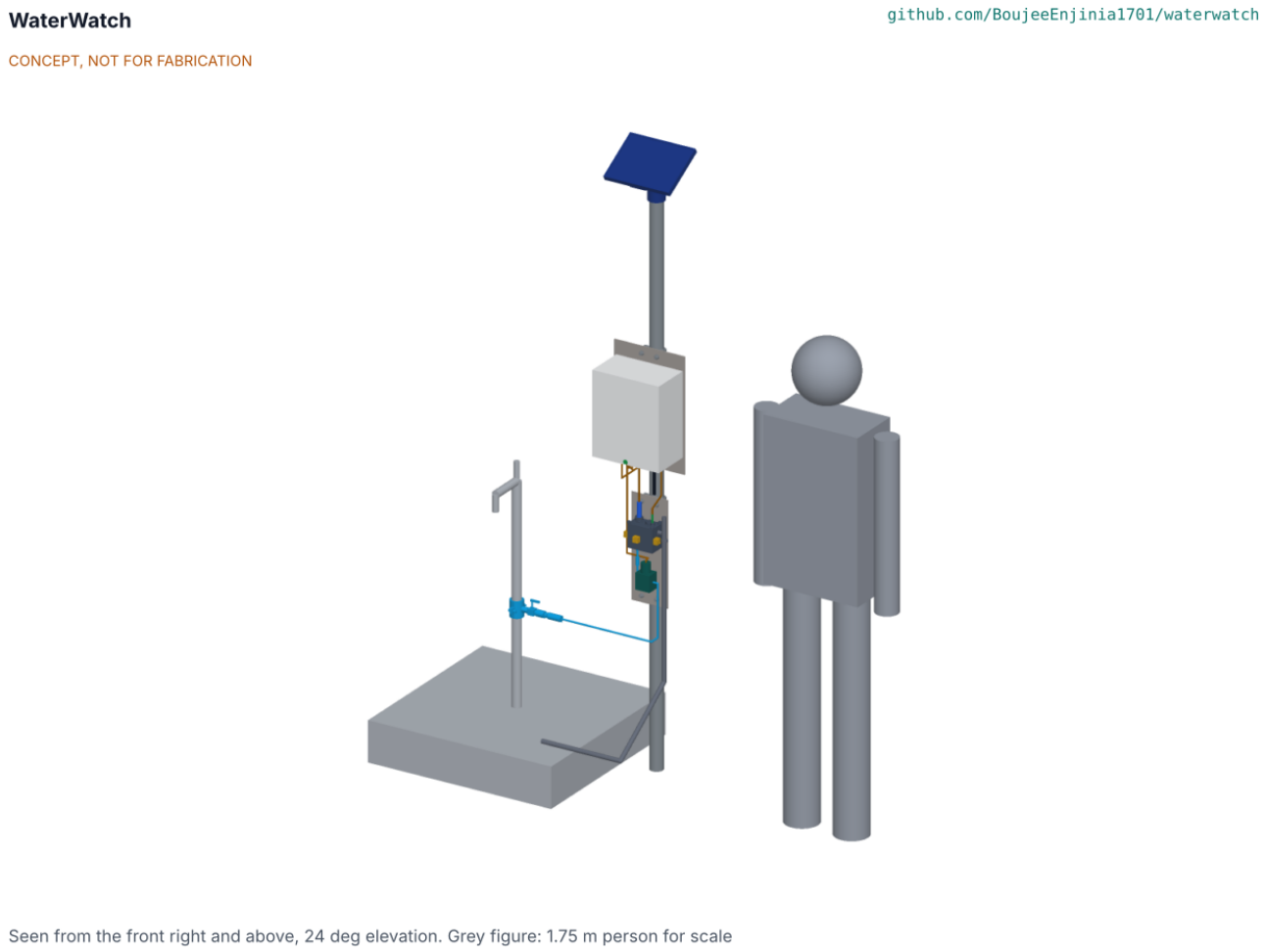


Figure 1. WaterWatch beside an existing tapstand (grey), with a 1.75 m person for scale. Solar panel on top, electronics enclosure under a white sun shield at eye height, flow-through cell below it, latching valve and sample line from a tee on the tapstand riser, drain hose to the tapstand basin. Rendered from the parametric model `cad/src/model.py`.

How it works

1. **Sample.** A saddle tee with an isolation valve on the tapstand riser feeds a strainer, a check valve, a pressure-compensating flow regulator (0.5 L/min) and a normally closed, direct-acting latching solenoid valve, with a second check valve at the cell inlet (WWT-DEC-001 F8). Where the supply is below the regulator's 0.7 bar, the flush is set to 110 s at commissioning (F5). Every hour the controller opens the valve for 90 s, flushing 0.75 L, about four volumes of the 0.19 L cell, then closes it. The sample enters at the bottom of the cell and leaves near the top, so the cell stays full and the electrodes stay wet between flushes. The outlet rises to an open air-break vent before the drain hose falls to the basin, so the hose cannot siphon the cell empty and the drain can never connect back to the cell.
2. **Measure.** During the last 10 s of flow the controller reads the chlorine sensor, because amperometric sensors need steady flow across the electrode. After the valve closes it waits 30 s for bubbles to rise clear, then reads turbidity for 5 s and temperature. The cell is opaque so daylight does not reach the turbidity detector.

3. **Log.** Each reading is stored with a timestamp on a microSD card (90 days or more) and in a short ring buffer in flash.
4. **Report.** Every 4 h the modem wakes, attaches to LTE-M, NB-IoT or 2G, and posts a compact batch to an open endpoint the operator chooses. If a reading crosses a threshold, the controller takes a confirming reading 15 min later rather than waiting for the next hourly one (WWT-DDR-002). If that one also crosses, it sends an SMS at once, retrying up to three times at 3 min spacing, without waiting for the next upload.
5. **Power.** A 5 W panel charges a 3.2 V, 6 Ah LiFePO4 battery through a small solar charger with temperature cut-offs. A ventilated sun shield keeps the enclosure cool enough for the battery to charge on hot days. Everything else sleeps between readings.

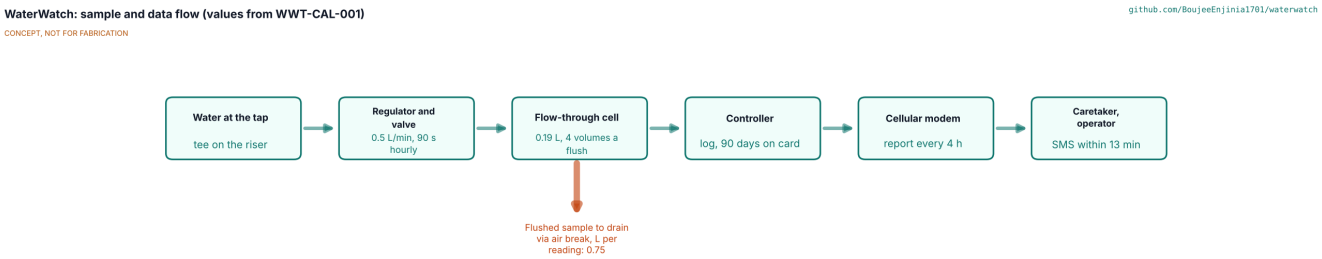


Figure 2. Sample and data flow. Values from WWT-CAL-001 for the default hourly schedule.

Main components

Table 1. Main components. Numbers match `bom/bom.csv` and Figure 5.

#	Component	Choice	Notes
1	Mounting pole, U-bolts and V-saddles	48.3 x 3.2 mm galvanized tube, 2.1 m above ground, 0.6 m in a 300 mm concrete footing beside the tapstand; four M8 U-bolts with V-saddles hold the two back plates (item 18)	Keeps the tapstand itself unchanged; factor 4.0 in a 35 m/s gust on the sun shield's area, footing factor 2.2 in saturated sand (WWT-CAL-001, I)
2	Solar panel	5 W monocrystalline, about 250 x 190 mm, tilted 30° toward the equator on a bought pole-top tilt mount	Oversized for the load so cloudy weeks are covered
3	Enclosure base	IP66 polycarbonate, 200 x 120 x 260 mm, UV-stabilized, pressure-equalizing vent; every cable, the vent and the antenna enter through the bottom face (WWT-DDR-003)	Light grey, under the sun shield (item 17)
4	Enclosure lid and gasket	Supplied with item 3; tamper-resistant screws	
5	Battery	LiFePO4, one 32700 cell, 3.2 V, 6 Ah (19.2 Wh), protection board and NTC (WWT-DEC-001 F1)	Chemistry chosen for heat tolerance and thermal stability
6	Controller board	ESP32-S3 module, LiFePO4 solar charger, LMP91000-class potentiostat front end, valve driver with boost to 12 V, microSD, real-time clock	Carrier board; a custom PCB is TRL 4 work
7	Cellular modem	LTE-M and NB-IoT with 2G fallback (SIM7000G class), SIM	Swappable; LoRaWAN module as a variant (WWT-DDR-001, D3)
8	Antenna	External LTE and GSM whip on a bulkhead in the bottom face, pointing down	Moved from the top so the sun shield can slide off (WWT-DDR-003)
9	Flow-through cell	Opaque black PVC or printed ASA block, 96 x 60 x 74 mm with a 14 mm back wall screwed to the cell plate, gasket and 10 mm lid on four studs with thumb nuts, 80 x 38 x 66 mm cavity, 0.19 L of water, baffle on the back wall, inlet in the floor and outlet near the top; lid glands for the sensors and a spare M20 port with a blanking plug	Half the TRL 2 volume so the flush clears it; opens without tools for cleaning; the spare port takes the pH probe variant (WWT-DDR-002)
10	Turbidity optics	860 nm infrared LED on the left wall, light-to-frequency detectors on the front wall (90°, scatter) and right wall (180°, reference), each in a small holder with a window, after Kelley et al. (2014)	ISO 7027 style wavelength
11	Free chlorine sensor	Membrane-free three-electrode sensor: graphite working electrode, Ag/AgCl reference, stainless counter, in a PVC holder, after Pan et al. (2015)	Decided (WWT-DDR-001, D1); long-term drift unknown
12	Temperature probe	DS18B20 in a stainless sheath	Also used for chlorine sensor temperature compensation
13	Latching solenoid valve	12 V, 1/4 in (6.35 mm) port, normally closed, direct acting, rated to open against at least 4 bar (F6), food-safe wetted parts; screwed to the cell plate under the cell	Latching, so it draws power only while switching; direct acting so it opens at 0.5 bar
14	Sample line		

#	Component	Choice	Notes
		Saddle tee clamp with isolation ball valve, strainer, check valve, pressure-compensating 0.5 L/min regulator, 1/4 in food-grade PE tube, and a second, inline check valve at the cell inlet (F8)	The only change to the tapstand; the regulator holds R8 from 0.7 to 4 bar
15	Drain hose and air break	Outlet elbow with an open air-break vent, then 12 mm hose to the tapstand basin	Basin by default, soakaway or container per site (WWT-DDR-001, D7)
16	Cables, glands and fuse	Sensor and panel leads, IP68 glands, battery fuse	
17	Sun shield	White aluminum hood with a 25 mm ventilated gap over the enclosure top, sides and front; flanges at the back held by four thumb screws; slides off forward	Added at TRL 3 so the battery can charge on hot days (WWT-CAL-001, C)
18	Mounting plates	Enclosure back plate 270 x 420 mm and cell back plate 140 x 385 mm, 3 mm aluminium, drilled	Added for construction (WWT-DDR-003)
19	Lugs and internal plate	The enclosure maker's four mounting lugs; a printed internal plate on the box's moulded bosses carrying the battery, controller and modem	Added for construction (WWT-DDR-003)
20	Fixings	Stainless screws, thumb screws, standoffs, battery tray screws, cable ties	Added for construction (WWT-DDR-003)

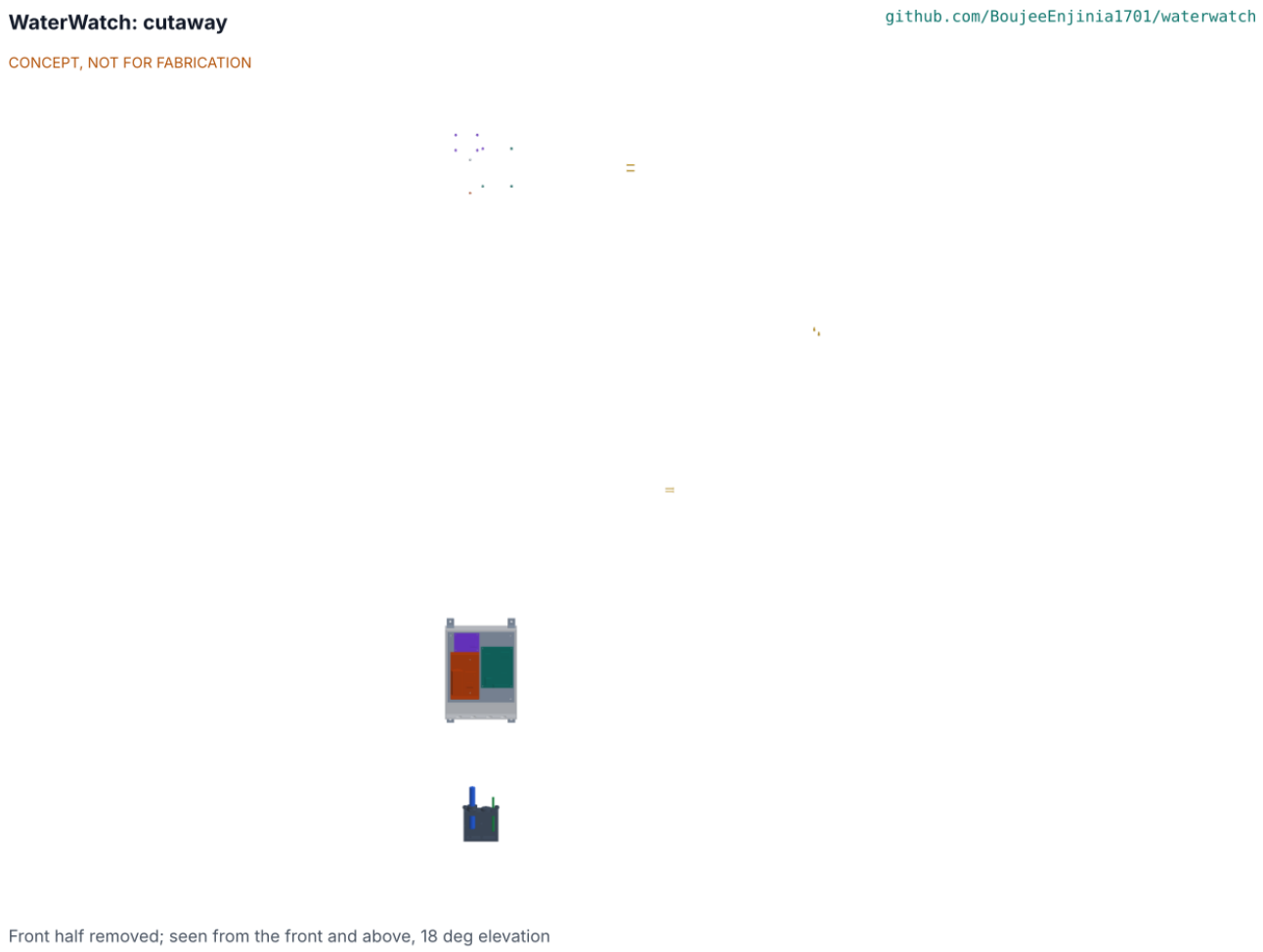


Figure 3. Section through the flow cell, 7 mm in front of its centre, looking at the front, with the sun shield removed. Top: enclosure with the modem (7, purple) and battery (5, orange) on the left and the controller board (6, teal) on the right, on the internal plate. Bottom: flow-through cell (9) with the chlorine sensor (11, blue) and temperature probe (12, green) entering through the lid, the LED and 180° detector holders of the turbidity optics (10, yellow) on the left and right walls, and the outlet near the top on the right.

Key numbers

All values come from WWT-CAL-001, which states its assumptions; the tags in brackets are the lines of `docs/04-calcs/sizing.py` that print them. The main assumptions are hourly readings, 130 s awake per reading at 0.264 W, a 50 µA sleep floor, six uploads a day at 20 mWh each, a 1.5 margin and 4.5 peak sun hours with a 0.6 derating.

Table 2. Key numbers.

Quantity	Value	Basis	Requirement
Daily energy with margin	0.540 Wh/day	0.229 Wh measuring, 0.120 Wh uploads, 9 mWh sleep and valve, 2 mWh status light, x 1.5 [A3]	
Autonomy without sun	28.5 days (24.2 at 0 °C)	15.36 Wh usable [B1]	R7 met on paper
Solar yield, 5 W panel	13.5 Wh per clear day, 7.5 Wh overcast	[B2]; 1.19 clear days to refill [B3]	R7 met on paper
Enclosure peak on a 45 °C day	50.8 °C with the sun shield; 69.8 °C without	Dusty enclosure, sun square to the front [C4]	R9 at risk
Flow-through cell	0.185 L of water	Parametric model [D1]	
Water per reading and per day	0.75 L; 18.0 L (19.8 L at regulator tolerance)	0.5 L/min for 90 s, 24 readings [D2], [D5]	R8 met on paper, thin margin
Old water left at the chlorine reading	2.9 %	Well-mixed cell [D3]	R1
Chlorine error at 0.5 mg/L, pH 7.5	±0.131 mg/L before drift, against ±0.2 mg/L	Site pH ±0.2 between visits, DPD reference [E4]	R1 at risk (drift unknown)
Room for sensor drift	0.15 mg/L or more	R1 restated above 1.0 mg/L (O5) with the shared photometer (O2) and the pH probe rule, 25 °C [E12]	R1 met on paper; R10 at risk
Data per month	0.36 MB	780 B payload; TLS session resumption, about 1.2 kB overhead (F7); 1.22 MB with a full handshake every time [G1, G3]	R6 met on paper
Confirmation to SMS	13.0 min worst case; 77 min from the event with a first-try SMS, 92 min with all three tries	Three tries at 3 min spacing [H1]; 15 min confirming reading [H2, H4]	R5 met on paper
Installation day	65 min critical path	Footing cast on the survey visit [J2]	R11 met on paper
Parts cost	$325 per base unit; $385 to $425 at a pH probe site	`bom/bom.csv` [K1], [K2]	R12: USD 25 over the USD 300 value-engineering target
Shared calibration kit	about $60 for the pH comparator and turbidity standards (estimate); handheld DPD photometer USD 379.99 to 599.99 (Hanna HI97711, WWT-MKT-001)	Shared per operator; the photometer is the monthly chlorine reference (WWT-DEC-001 O2, decided 2026-10-25)	Not in the unit cost

The chlorine reading is the least certain number. Free chlorine is the sum of hypochlorous acid (HOCl) and hypochlorite (OCl^-); their split depends on pH, with a pKa of 7.54 at 25 °C, so at pH 8 only 0.26 of free chlorine is HOCl, while at pH 7 0.77 is [E2]. Amperometric sensors respond mainly to HOCl, so the same free chlorine gives very different signals across the pH range in R1. Calibrating at the site against the DPD comparator absorbs the pH on the day of calibration, but a change of 0.2 pH before the next visit shifts the reading by 12 % at pH 7.0, 22 % at

pH 7.5 and 38 % at pH 8.0 [E4]. The site pH method is therefore adequate at near-neutral, stable sites and not elsewhere. That is why WWT-DDR-002 fits the pH probe variant where site pH is above 7.5 or moves more than 0.2 between visits, and relaxes R1 to ±0.2 mg/L or ±25 %, since the DPD comparator used as the field reference already takes ±0.07 mg/L. With both, all eight cases in WWT-CAL-001 meet R1 before drift [E6]; at 40 °C, 1.5 mg/L at pH 7.5 with site pH misses by 0.006 mg/L [E7].

Key design choices

Each choice below was decided by Amish on 2026-09-25, going with the recommendation (WWT-DDR-001 and WWT-DDR-002). The options considered are kept for the record.

- **Chlorine sensor (D1).** Chosen: a membrane-free graphite three-electrode sensor with a potentiostat front end, after Pan et al. (2015) and the pencil graphite work in PLOS ONE (2021), about $25 in parts, with a monthly DPD comparison built into the maintenance routine. Not chosen: an industrial membrane amperometric probe, well proven but about $1,900 for the probe alone (Sensorex FCL); an ORP probe as a proxy, about $175 as a kit (Atlas Scientific), which tracks chlorine only loosely. The long-term performance literature (Water Science and Technology, 2025) could not be checked in full; see `docs/REVIEW.md`.
- **pH (D2).** Chosen: site pH entered at installation and checked monthly, at no cost. A pH probe and interface (about $60 to $100, estimate) is a documented variant. Under WWT-DDR-002 it is fitted at sites whose pH is above 7.5 or moves more than 0.2 between visits, in a spare port in the cell lid that is plugged at other sites; it is priced per variant site, outside the $300 base-unit budget, which stays unchanged.
- **R1 target (WWT-DDR-002).** Chosen: relax to ±0.2 mg/L or ±25 %, whichever is greater. Not chosen: keep ±0.1 mg/L and accept that R1 is not met; keep it and make the pH probe standard, which exceeds the budget. Restated by Amish on 2026-10-25 as ±30 % above 1.0 mg/L (WWT-DEC-001 O5), with a shared DPD photometer as the monthly reference (O2).
- **Confirming reading (WWT-DDR-002).** A firmware rule: after a first threshold crossing, the next reading is taken 15 min later instead of at the next hour, cutting the worst case from event to SMS from 122 min to 77 min for about 10 mWh and 0.75 L per event.
- **Footing on the survey visit (WWT-DDR-002).** The pole footing is dug and cast, with the pole set, on the survey visit, so installation day takes about 65 min instead of 140 min.
- **Sampling by timed flush (D5).** An hourly 90 s flush uses 18.0 L per day; continuous flow at 0.5 L/min would use 720 L. At TRL 3 the cell was halved in volume, the fixed restrictor replaced by a pressure-compensating regulator, and the outlet moved near the top with an air break, all within this decision.
- **Pole beside the tapstand (D4).** A separate pole keeps the panel above head height, keeps the electronics out of reach of splashing and buckets, and needs no drilling into the tapstand. The cost is one more concrete footing, which also sets the installation time (R11).
- **Connectivity (D3).** Cellular LTE-M and NB-IoT with 2G fallback and SMS alerts for the first build, with the modem as a swappable module so LoRaWAN is a variant; this keeps the "GSM or LoRa" pitch.
- **LiFePO4 battery (D9).** Chosen over lithium-ion 18650 cells for heat tolerance and thermal stability, with charging blocked below 0 °C and above 45 °C. At TRL 3 a ventilated sun shield was added because, without it, the enclosure stays above 45 °C for almost all of a clear hot day.
- **Default alert thresholds (D6).** Free chlorine below 0.2 mg/L and turbidity above 5 NTU, confirmed by two consecutive readings, adjustable per site.
- **Drain (D7).** To the tapstand basin by default, to a soakaway or a container for non-drinking use where users prefer, decided per site.

- **First site type (D8).** Tapstands first; tank outlets and kiosks later.
- **SwapCell (D10).** Not used: WaterWatch needs about 0.5 Wh per day, and the portfolio's 48 V SwapCell pack does not fit.

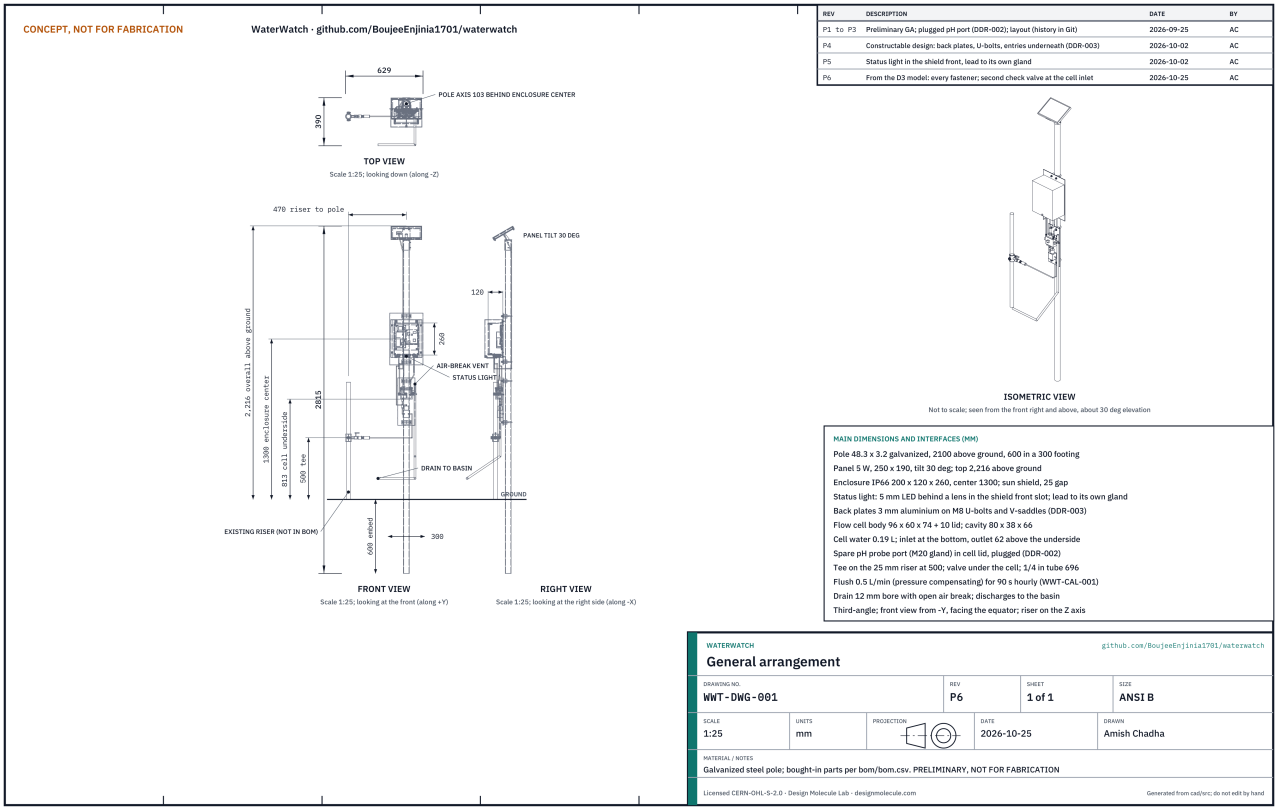


Figure 4. General arrangement WWT-DWG-001, Rev P4, generated from `cad/src/model.py` *by* `cad/src/sheets.py` *. Preliminary, not for fabrication.*

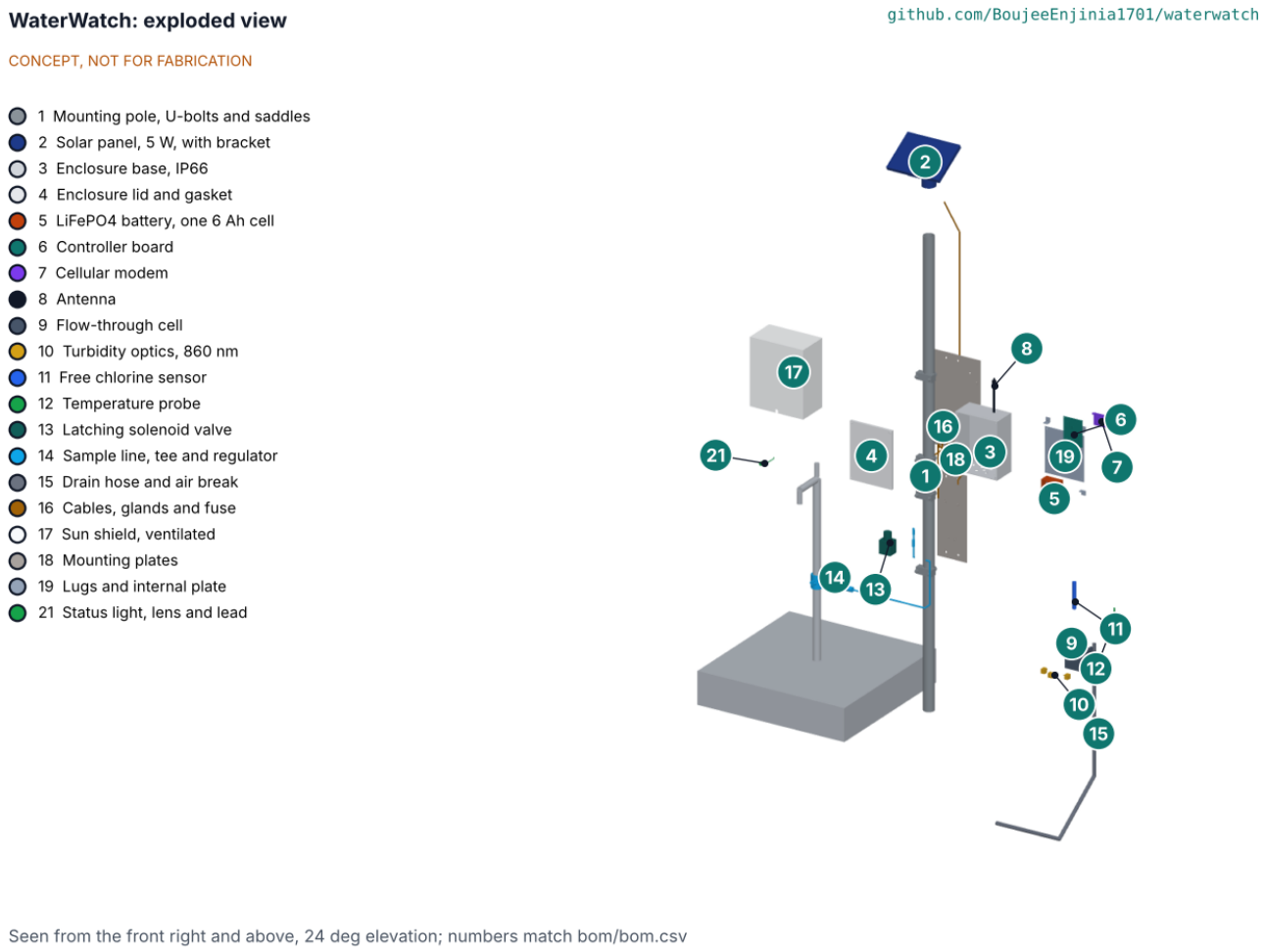


Figure 5. Exploded view with BOM numbers. The tapstand (grey) is existing infrastructure and not in the BOM.

Safety

Safety: WaterWatch is a monitoring aid, not a certification of water safety. A normal reading does not mean the water is free of pathogens or chemical contaminants; accredited laboratory testing, including microbiological testing, remains necessary. Alerts must be worded as measurements, never as "safe" or "unsafe".

Safety: The design contains a lithium iron phosphate battery (19.2 Wh). Use cells with a protection board, a fuse at the battery, and charging cut-offs below 0 °C and above 45 °C. Keep the enclosure vented, light-colored and under its sun shield; WWT-CAL-001 shows that an unshaded enclosure reaches about 70 °C inside on a 45 °C day.

Safety: The sample line connects to a pressurized drinking water supply. Fit an isolation valve and both check valves (the supply train and the cell inlet) so nothing can flow back into the supply, even when the main runs dry, use food-safe wetted materials, keep the open air break at the cell outlet, and never let the flushed

sample return to the drinking water. Fitting the saddle tee needs the supply shut off unless a tool for tapping under pressure is used. Flushed water contains chlorine; drain it where it cannot pool against the tapstand or be mistaken for a separate water point.

Safety: Calibration and cleaning use chemicals: DPD reagent tablets, and formazin turbidity standards, which are hazardous if made from their precursors. Use prepared, stabilized standards, gloves and the supplier's safety data sheets; do not make formazin in the field.

- The pole is a fall and impact hazard if it is not set properly; set it in concrete and keep sharp edges on the panel frame and clamps covered.
- Radio and battery equipment must meet local type-approval rules before any field installation.

Open questions

- [ ] What does the published long-term data say about low-cost chlorine sensor drift, and is it below the 0.15 mg/L per month that R1 and R10 now allow? (the WST 2025 paper is still unchecked in full)
- [ ] Does the sun shield cut the solar gain as much as assumed (a quarter)? (R9)
- [ ] How fast do the cell windows and the chlorine electrode foul at a turbid site, and does a monthly clean suffice? (R2, R10)
- [ ] How much does source pH move between monthly visits at candidate sites? (R1)
- [ ] Confirm with caretakers and operators the alert and data defaults decided by Amish on 2026-10-02 (WWT-DEC-001): SMS in the local language to the caretaker and the operator's maintenance contact, and data owned by the scheme operator.
- [ ] Approach the first candidate partner, Water Mission, and choose the first region from its East African country programs (decided by Amish, 2026-10-02, WWT-DEC-001).